Homework 5

BME 5910 - Biomedical Optics

Garrett Kunkler

10/5/2026

Problem 3

```
% Givens
x_vec = linspace(-2,2,1000); % Define mesh spacing
[x,y] = meshgrid(x_vec); % Create square mesh
r = sqrt(x.*x + y.*y);
S_0 = 5;
FWHM_beam = 1;
R = FWHM_beam * sqrt(1/log(4))
```

```
R =
0.8493
```

```
S = S_0 * exp(-2*(r/R).^2);

% Assume same Green's function (but normalized to integrate to 1)
G = 1/pi/R^2 * exp(-2*(r/R).^2);

% Perform convolution
C = conv2(S,G, "same")*(x_vec(2)-x_vec(1))*(x_vec(2)-x_vec(1));

FWHM_S = get_FWHM(S,x_vec)
```

```
FWHM_S =
0.9970
```

```
FWHM_C = get_FWHM(C,x_vec)
```

```
FWHM_C =
1.4094
```

```


% Calculate FWHM
function fwhm = get_FWHM(f, x)

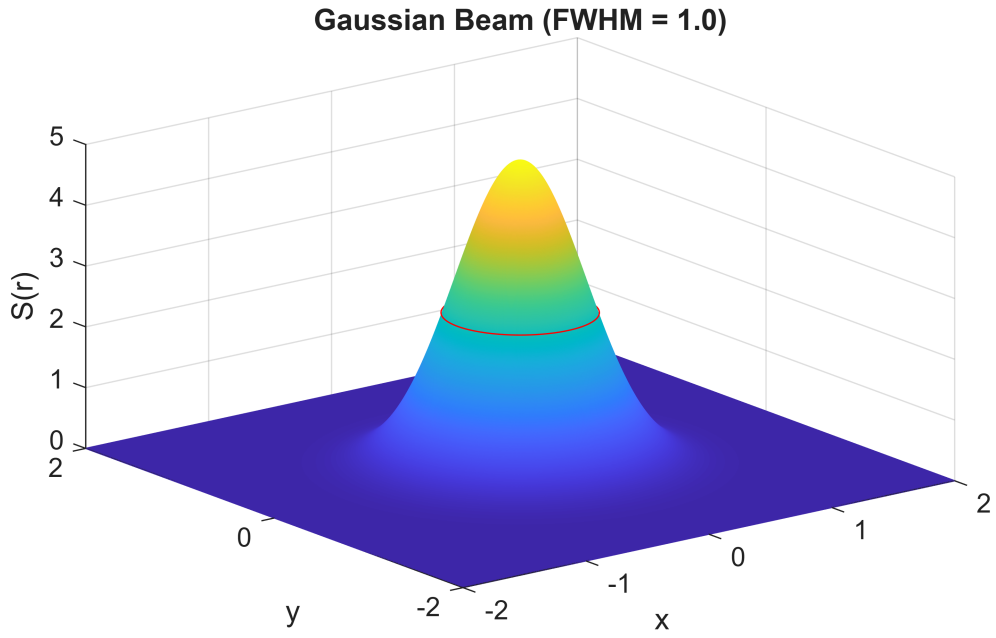
    [f_max_x,~] = max(f);
    [f_max,~] = max(f_max_x);

    i_low = find(f_max_x > 0.5 * f_max,1,'first');
    i_high = find(f_max_x > 0.5 * f_max,1,'last');

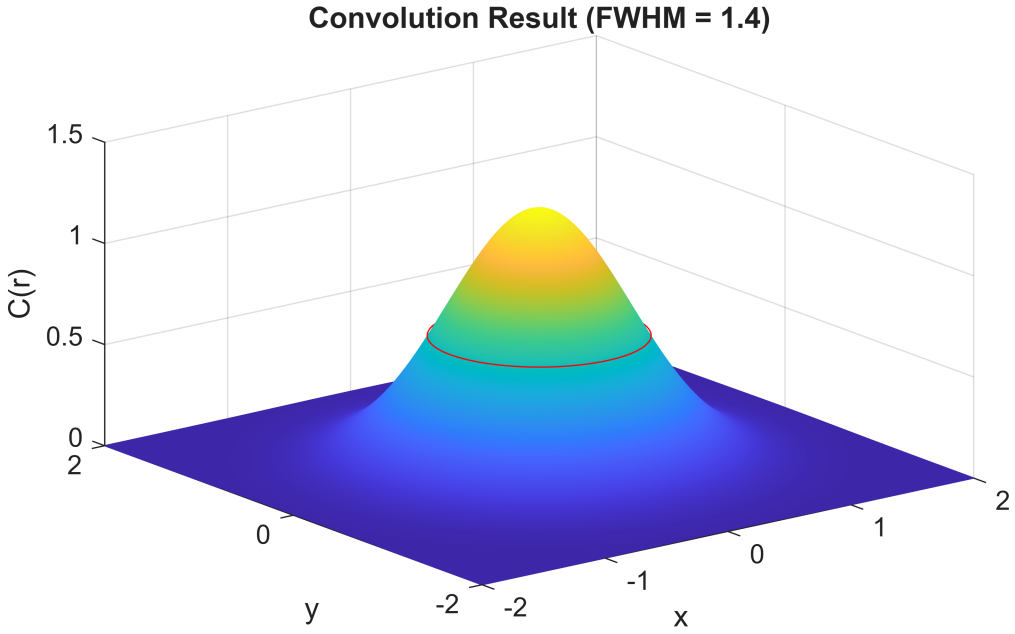
    fwhm = x(i_high) - x(i_low);

end
```

```
% Plot Original Gaussian S(r)
mesh(x,y,S)
hold on;
phi = linspace(0,2*pi, 100);
plot3(FWHM_S/1.95*cos(phi), FWHM_S/1.95*sin(phi), ones(1,100)*max(S,[], 'all')/2,
'r-')
hold off;
xlabel('x')
ylabel('y')
zlabel('S(r)')
title(gca, sprintf('Gaussian Beam (FWHM = %.1f)',FWHM_S))
```




```
% Plot Original Gaussian S(r)
mesh(x,y,C)
hold on;
plot3(FWHM_C/1.95*cos(phi), FWHM_C/1.95*sin(phi), ones(1,100)*max(C,[], 'all')/2,
'r-')
hold off;
xlabel('x')
ylabel('y')
zlabel('C(r)')
title(gca, sprintf('Convolution Result (FWHM = %.1f)',FWHM_C))
```


Convolution Result (FWHM = 1.4)
C(r)
1.5
1
0.5
0
2
0
y
-2
-2
-1
0
1
2
x